

The V'_3 -lemma, reduced

Cosine-grid mesh, Chebyshev source formula, the $\mathcal{A}_{15}$ neighbourhood, and the coefficient box G_ε

Stefan Hamann

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Abstract

Round 379 attacks the named remainder V'_3 of [1] (the last fourteen Prüfer steps of v_2 at the $+x$ mask lie in $\mathcal{A}_{15}$). V'_3 itself is neither proved nor refuted in the construction. What is proved: the smooth-border nodes are the cosine grid $x = \cos(2\pi k/L)$ with $L = 4h - 2$ (construction identity $M = 2h$, $L = 2M - 2$); at the $+x$ 20% mask consecutive bulk nodes are equidistant in $\arccos$ (fold-gap constant; ratio 1 on all 181 windows); equal weights on the occupied cosine nodes reproduce Chebyshev coefficients ($\gamma_1 = \frac{1}{2}$, $\gamma_k = \frac{1}{4}$ for $k \geq 2$); the Fejér factor $\sin^2(\theta/2)$ of the fold is Jacobi-(0, 1) with $|\gamma_n - \frac{1}{4}| = 1/(4(2n+1)^2)$; two-period profiles about $\pi/2$ with amplitude $a \leq 0.85$ lie in $\mathcal{A}_{15}$, while a slow-then-fast *block* with the same extremes need not; a coherent last-12 γ -scale of 1.5 stays in $\mathcal{A}_{15}$ and a scale of 2.0 leaves it (the actual v_2 -violator of [1] sits at scale 6.5–8). The hydra-legal reduction is the coefficient box G_ε : the last twelve Jacobi γ_k of the v_2 -chain satisfy $|\gamma_k - \frac{1}{4}| \leq \frac{1}{16}$ and $|\log(\gamma_{k+1}/\gamma_k)| \leq \frac{2}{5}$. On the 181-window surface G_ε holds (max last-3 0.0375, max last-12 0.0519, max jump 0.337) and V'_3 holds (0 remainder-violators). Klein-gap and χ windows are not exceptions. The chain $T'_2 \Leftarrow \dots \Leftarrow V_2 \Leftarrow V'_3 \Leftarrow G_\varepsilon$ stands, with G_ε the remaining named lemma. No RH claim.

Status. Lemmas 1–2 are proved (construction identities; Jacobi closed form). Lemma 3 is a machine-checked identity on the occupied cosine grid. Lemmas 4–5 are finite statements about step sequences and Jacobi rays. Proposition 1 names G_ε and records Path A on 181 windows. The hydra test of [2] is met: G_ε is strictly more elementary than V'_3 . No RH claim.

1 Recollection

V'_3 of [1] §5: the last fourteen spatial steps of the degree-accumulated Jacobi–Prüfer phase of v_2 on the last fifteen bulk-border nodes at the $+x$ 20% mask lie in the region $\mathcal{A}_{15} \subset \mathbb{R}^{14}$ of sequences that are V_2 -regular for every incoming remainder $\eta \in [0, \pi)$. The Wronskian step formula and the contrast dichotomy (constant steps never violate; a slow-then-fast block $(0.4^{21}, 2.8^3)$ does) are theorems of [1]. The Jacobi coefficients $(\hat{\alpha}_k, \gamma_k)$ do not depend on x ; the x -variation of the step is the Christoffel profile. The coefficients were *measured* near Chebyshev ($|\gamma - \frac{1}{4}| \leq 0.041$ on the last three of the pins). This note derives the mesh and the Chebyshev source formula, and isolates the remaining bound as G_ε . An $8 \times$ last-12 γ -scale on χ_3 $w = 9$ is a v_2 -violator [1]: abstract Jacobi-positivity is not enough, and the source geometry must enter.

2 The cosine-grid mesh

The fold of [3] §2 pushes lag-cells $j = 0, \dots, L - 1$ to

$$x = \cos(2\pi k/L), \quad k = \min(j, L - j).$$

The lag truncation is $M = 2h$ and the circle length is $L = 2M - 2$, hence $L = 4h - 2$. On FRAME-A, the chain depth is $N = h$, so the Chebyshev unit (degree $n = N - 2$ times one fold-step) is

$$n \cdot \frac{2\pi}{L} = \pi \frac{h-2}{2h-1} \xrightarrow{h \rightarrow \infty} \frac{\pi}{2}. \quad (1)$$

At $h = 184$ this is $182\pi/367 \approx 1.558$.

Lemma 1 (cosine grid). *Smooth-border nodes are a subset of the cosine grid $x = \cos(2\pi k/L)$ with $L = 4h - 2$. At the $+x$ 20% mask, consecutive bulk-border nodes are equidistant in arccos: the fold-gap is constant on the 15-node window.*

Proof. The identity $L = 2M - 2$ is the return of `build_rung`; $M = 2h$ is the half-filling lag truncation (asserted in the window builder, integer-checked). The fold map is $x = \cos(2\pi k/L)$ by construction. Equidistance in arccos on the mask window is the statement that the surviving fold indices form an arithmetic progression; it is gated on the 181-window surface (ratio of consecutive $\Delta\theta$ equal to 1 to machine precision). $\square$

Thus a Chebyshev polynomial T_n on this mesh has constant spatial steps equal to (1) (fold-gap 1) or twice that (fold-gap 2). Both values lie in $(\pi/3, \pi)$ for every admitted $h \geq 92$: no run of length ≥ 3 can form from a constant Chebyshev step, because $r \geq 3$ demands mean interior step $< \pi/3$ [1].

3 The γ -source formula

The folded weights carry an explicit Fejér factor $4 \sin^2(\theta/2) = 2(1-x)$. Orthogonal polynomials for the Jacobi weight $(1-x)$ on $[-1, 1]$ (parameters $(\alpha, \beta) = (0, 1)$) have monic recurrence coefficients

$$\gamma_n = \frac{n(n+1)}{(2n+1)^2}, \quad \left| \gamma_n - \frac{1}{4} \right| = \frac{1}{4(2n+1)^2}.$$

At $n = 182$ the right-hand side is $1.9 \cdot 10^{-6}$. The Fejér factor alone does not produce the measured $O(10^{-2})$ deviation.

Lemma 2 (Fejér Jacobi). *For the weight $(1-x)$ on $[-1, 1]$, $|\gamma_n - \frac{1}{4}| = 1/(4(2n+1)^2)$ exactly (rational identity).*

Lemma 3 (equal-weight cosine grid). *On the occupied cosine nodes of FRAME-A $w = 9$ (367 atoms) with equal weights, the monic Stieltjes coefficients satisfy $\gamma_1 = \frac{1}{2}$ and $|\gamma_k - \frac{1}{4}| < 10^{-5}$ for $2 \leq k \leq 40$ (one fold index of 368 is unoccupied; the residual is 10^3 times smaller than the weighted source). The node set alone is Chebyshev; the measured $O(10^{-2})$ is a weight perturbation.*

The source weights are the r342 dictionary (archimedean digamma $F_A(\xi) = -\log \pi + \Re \psi(\frac{1}{4} + i\xi/2)$ plus the exact two-cell tent of the prime layer, v_{pred} relative deviation $< 10^{-4}$) times the Fejér factor. The Jacobi coefficients of v_2 are therefore

$$\gamma_k = \frac{1}{4} + \delta_k^{\text{Fejér}} + \delta_k^{\text{darm}},$$

with $\delta^{\text{Fejér}}$ explicit and negligible at half-filling, and δ^{darm} the only live term. This is the source formula. It is not a bound: δ^{darm} is the named remainder below.

WKB at the mask $x = \frac{3}{5}$, for a constant γ with $|\gamma - \frac{1}{4}| \leq \varepsilon$, the spatial-step factor against Chebyshev is $\sqrt{1-x^2}/\sqrt{4\gamma-x^2}$. At $\varepsilon = \frac{1}{16}$ the WKB band is $[1.332, 2.012]$, inside the safe uniform band of Lemma 4. The source last-12 α_k satisfy $|\alpha| \leq 0.071$ (pins), so the turning points stay near ± 1 .

4 From bounds to $\mathcal{A}_{15}$

The obstruction of [1] is a *block*: a long slow run then three fast steps. An interleaved two-period oscillation with the same extremes does not violate.

Lemma 4 (two-period neighbourhood). *Let $\Delta = \pi/2$ and $d_i = \Delta(1+a(-1)^i)$ for $i = 0, \dots, 13$, with $0 \leq a \leq 0.85$ (all steps in $(0, \pi)$). Among 2001 remainders $\eta \in [0, \pi)$, there are 0 V_2 -violators. At $a = 0.85$ the step-ratio is 12.3 and the small step $0.236 < \pi/3$, so $r \geq 3$ dwells are possible: they do not produce a violator because slow and fast alternate.*

The companion script also gates the blocked rearrangement of the same values: a slow-then-fast block can leave $\mathcal{A}_{15}$ at step-ratio 1.83 (G4 of [1], refined). Ratio is not the obstruction; spatial segregation is.

Lemma 5 (dangerous ray). *On χ_3 $w = 9$, multiplying the last twelve γ_k by a coherent scale s : $s = 1.5$ has 0 $\mathcal{A}_{15}$ -violators (401 remainders); $s = 2.0$ has 2/401; the actual v_2 occupancy first fails at $s = 6.5$ (the $8\times$ of [1] is past the threshold). Independent jitter of the last twelve inside $|\gamma - \frac{1}{4}| \leq 0.06$ (40 draws) never leaves $\mathcal{A}_{15}$.*

The source lives at relative scale ≤ 1.24 about $\frac{1}{4}$ (last-12 $|\gamma - \frac{1}{4}| \leq 0.052$). The first $\mathcal{A}_{15}$ failure on the coherent ray is at scale 2, four times farther in the ε -metric (0.25 versus 0.06). Abstract Jacobi-positivity remains insufficient (the ray does leave $\mathcal{A}_{15}$); the source box is far inside the first failure.

5 The smaller lemma G_ε

Definition 1 (G_ε). *Let $n = N - 2$ be the degree of v_2 and let $(\gamma_k)_{k=1}^n$ be the monic Jacobi coefficients of the bordered chain of $\mu - \nu$. Condition G_ε is that the last twelve coefficients satisfy*

$$|\gamma_k - \frac{1}{4}| \leq \frac{1}{16} \quad \text{and} \quad |\log(\gamma_{k+1}/\gamma_k)| \leq \frac{2}{5}.$$

Proposition 1 (reduction). *G_ε is strictly more elementary than V'_3 in the sense of the hydra ward of [2] §5:*

1. Finite. Twelve real inequalities on the chain coefficients. V'_3 is a point of $\mathbb{R}^{14}$ together with a compact scan of $\eta \in [0, \pi)$.
2. Closer to the construction. The coordinates are the three-term recurrence of $\mu - \nu$ (moments, r342 dictionary, cosine-grid nodes). No Prüfer phase, no Wronskian occupancy, no mask remainder η .
3. Fewer quantifiers. No $\forall\eta$.

On the 181-window surface (FRAME-A 89, FRAME-B 8, χ_3 42, χ_4 42): G_ε holds, the mask mesh has $\Delta\theta$ -ratio 1, and the last-14 step tuple lies in $\mathcal{A}_{15}$ (0 remainder-violators among 401 remainders per window; 2001 on the pins). Klein-gap windows kz 51, 75, 79, 103, 111 have last-3 $|\gamma - \frac{1}{4}| \leq 0.0062$ (smaller than the bulk small- N worst kz 19 at 0.0375). χ_3 index 16 produces remainder-(1, 1, 1) tails, all V_2 -regular. The chain is

$$T'_2 \Leftarrow g_i \geq \frac{3}{8} \Leftarrow \text{MED-CAP} \Leftarrow X_n \Leftarrow V_2 \Leftarrow V'_3 \Leftarrow G_\varepsilon.$$

G_ε is not a Chebyshev-asymptotic toward L^* : it is a finite box at half-filling, using the cosine-grid identity rather than Stahl–Totik regularity. The equal-weight case (Lemma 3) shows the box is a statement about d_{arm} , i.e. about the tent/digamma dictionary, not about the node set.

6 Adversaries

A coherent last-12 scale of 2 leaves $\mathcal{A}_{15}$ for some η (Lemma 5): V'_3 uses the source. Klein-gap windows (r329/r355 families) sit in a *tighter* coefficient box, not a looser one. χ worlds are not exceptions (χ_3 carries the v_2 -triple, regular; χ_4 last-12 max 0.0519, inside $\frac{1}{16}$). Depth improves the bound (N= 1721 last-3 0.0011). A weight-cluster at the mask still does not fire [1].

7 What remains

To prove G_ε cofinally: bound δ_k^{darm} for $k \in [n - 12, n)$ from the closed-form lag density (r342), uniformly in the window. The measured law is that the deviation is largest at small N and shrinks; the 181-window surface includes every currently admitted small- N row. The implication $G_\varepsilon + \text{mesh} \Rightarrow V'_3$ is a Jacobi-on-cosine-grid statement, gated on the 181-tuple and on the coherent ray through scale 1.5, not an exhaustive 12-dimensional box scan. T1 / C_K unchanged. No RH claim.

8 Machine checks

Every numbered lemma is re-proved in `verify_v3prime.py` (same directory). Standalone: $L = 4h - 2$; Jacobi-(0, 1) identity over $\mathbb{Q}$; two-period $\mathcal{A}_{15}$; contrast regression; WKB band. Construction: mask $\Delta\theta$ -ratio, G_ε and Path A on the pins, equal-weight Chebyshev, the coherent ray, kz46 (step-ratio 5.85, still in $\mathcal{A}_{15}$). Optional `--full`: the 181-window census. Exit line: V3PRIME LEMMA VERIFIED.

Claim boundary

Finite identities on the cosine grid and the Fejér Jacobi coefficients, a finite two-period neighbourhood of $\mathcal{A}_{15}$, a named coefficient box G_ε to which V'_3 reduces, and a 181-window census. No statement about zeros of $\zeta(s)$, in either direction. No RH claim.

References

- [1] S. Hamann, *The V_2 -lemma, reduced*, standalone note, August 2026, `rh/problem/v2_lemma_v3.tex`.
- [2] S. Hamann, *Run-length regularity V_2 at the x -mask*, standalone note, August 2026, `rh/problem/v2_regularity.tex`.
- [3] S. Hamann, *The length invariant X_n of a folded sign field*, standalone note, August 2026, `rh/problem/xn_invariant.tex`.